CISS245: Advanced Programming
Quiz q2501

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Q1. You want to write a random library that allows you to do this:

```
int i = randrange(5);           // i is a random integer in {0, 1, 2, 3, 4}
int j = randrange(3, 9);        // j is a random integer in {3, 4, 5, 6, 7, 8}
double x = randrange(5.0);      // x is a random double in interval [0.0, 5.0]
double y = randrange(2.0, 5.0); // y is a random double in interval [2.0, 5.0]
```

Write the header file below.
Answer:

```
// file: random.h
#ifndef RANDOM_H
#define RANDOM_H

int randrange(int end);
int randrange(int start, int end);
double randrange(double end);
double randrange(double start, double end);
void init_random();

#endif // RANDOM_H
```

Q2. Continuing the above question, implement the functions in above header file.
Answer:

```
// file: random.cpp
void init_random()
{
    std::srand((unsigned int) time(NULL));
}

int randrange(int end)
{
    return std::rand() % end;
}

int randrange(int start, int end) {
    return start + (std::rand() % (end - start));
}

double randrange(double end) {
    return (double(rand()) % end);
}
```

```
double randrange(double start, double end) {
    return start + double(std::rand()) % (end - start);
}
```

Instructions

In the file `thispreamble.tex` look for

```
\renewcommand\AUTHOR{}
```

and enter your email address:

```
\renewcommand\AUTHOR{jdoe5@cougars.ccis.edu}
```

(This is not really necessary since alex will change that for you when you execute `make`.) In your bash shell, execute "`make`" to recompile `main.pdf`. Execute "`make v`" to view `main.pdf`.

Enter your answers in `main.tex`. In the bash shell, execute "`make`" to recompile `main.pdf`. Execute "`make v`" to view `main.pdf`.

For each question, you'll see boxes for you to fill. For small boxes, if you see

```
1 + 1 = \answerbox{}.
```

you do this:

```
1 + 1 = \answerbox{2}.
```

`answerbox` will also appear in "true/false" and "multiple-choice" questions.

For longer answers that need typewriter font, if you see

```
Write a C++ statement that declares an integer variable name x.
\begin{answercode}
\end{answercode}
```

you do this:

```
Write a C++ statement that declares an integer variable name x.
\begin{answercode}
int x;
\end{answercode}
```

`answercode` will appear in questions asking for code, algorithm, and program output. In this case, indentation and spacing is significant. For program output, I do look at spaces and newlines.

For long answers (not in typewriter font) if you see

```
What is the color of the sky?
\begin{answerlong}
\end{answerlong}
```

you can write

```
What is the color of the sky?
\begin{answerlong}
The color of the sky is blue.
\end{answerlong}
```

A question that begins with "T or F or M" requires you to identify whether it is true or false, or meaningless. "Meaningless" means something's wrong with the question and it is not well-defined. Something like "$1 + 2 = 4$" is either true or false (of course it's false). Something like "$1+_2 = 4$?" does not make sense.

When writing results of computations, make sure it's simplified. For instance write 2 instead of $1 + 1$.

Higher level classes.

For students beyond 245: You can put LaTeX commands in `answerlong`.

More examples of meaningless statements: Questions such as "Is $42 = 1+_2$ true or false?" or "Is $42 = \{2\}^{\{3\}}$ true or false?" does not make sense. "Is $P(42) = \{42\}$ true or false?" is meaningless because $P(X)$ is only defined if X is a set. For "Is $1 + 2 + 3$ true or false?", "$1 + 2 + 3$" is well–defined but as a "numerical expression", not as a "proposition", i.e., it cannot be true or false. Therefore "Is $1 + 2 + 3$ true or false?" is also not a well-defined question.

More examples of simplification: When you write down sets, if the answer is $\{1\}$, do not write $\{1, 1\}$. And when the values can be ordered, write the elements of the set in ascending order. When writing polynomials, begin with the highest degree term.

When writing a counterexample, always write the simplest.